

SIMATS ENGINEERING

ASSESSMENT TOOL 3

COURSE NAME: INTERNET PROGRAMMING

COURSE CODE: CSA4305

COURSE OUTCOME COVERED

CO 4: Formulate data-driven web solutions using XML, XSLT, and JSP within the MVC framework. (L4)

Assessment Tool: Class Test

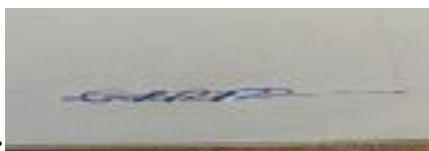
Weightage: 10%

QUESTIONS: Total Marks: 20

1. Explain the fundamental concepts used in web development by defining the following terms with suitable explanations: **XML, JSP, MVC architecture, XSLT, XML Schema, JSTL, PHP, Database Connectivity, and Parser**. Your answer should clearly describe the purpose and role of each concept in web applications.
2. Explain how the above technologies are interconnected in building a web application. Describe how XML is structured and validated, how JSP and PHP are used for server-side processing, how MVC architecture organizes the application, and how database connectivity and parsers contribute to data processing and communication.

Code of Conduct:

I G.Pradeep ranga reddy 192311289(Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.



Signature of the student:

RUBRICS FOR CALCULATION:

Criteria	Weightage	Level 4 – Exemplary	Level 3 – Proficient	Level 2 – Developing	Level 1 – Beginning
Concept Understanding	25%	Clearly explains all concepts with complete accuracy and depth	Minor gaps in understanding	Basic understanding with some errors	Incorrect or very limited understanding

Content Coverage	25%	Covers all required topics comprehensively	Covers most topics with minor omissions	Covers some topics only	Very few topics covered
Technical Accuracy	20%	All explanations are technically correct and precise	Minor technical errors	Some incorrect or vague explanations	Major technical errors
Integration of Concepts	15%	Effectively connects and relates concepts logically	Some connections made with minor gaps	Limited connections between concepts	No clear relationship between concepts
Clarity & Presentation	15%	Well-structured, clear, and easy to understand	Mostly clear with minor issues	Somewhat unclear or poorly organized	Poorly structured and difficult to understand

CO4-AT3.

1) Fundamental concepts in web development

1) XML.

XML is a markup language used to store, organize and transport data in structured format. It allows developers to define their own tags.

```
<student>
```

```
<name>Pradeep</name>
```

```
<age>22</age>
```

```
</student>
```

2) JSP.

JSP is a Java-based server-side of technology used to create dynamic web pages to contain Java/JSTL-based logic.

Purpose:

JSP and HTML generates the response.

Example: login pages, displaying database records, user dashboards, etc.

3) mvc architecture.

mvc stands for models - view - controllers.

It divides a web application into three parts:

- model: Handles data and business logic, such as data base operations.
- view: Handles the user interface, such as HTML, JSP Pages.
- controller: Receives user requests and controls the flow between model and view.

4) XSLT.

XSLT schema, commonly called xsl.

XSLT is a language used to transform XML documents into other formats, such as HTML, XML or text.

6) JSTL.

JSTL is a collection standard tags u
JSP Pages. It provides tags for
common operations such as.

- Conditions.
- Loops
- Formatting with XML
- Working with XML.
- Database-related operations.

7) PHP

PHP is a server-side scripting
language mainly developing dynamic
web applications.

PHP can:

- Process form submissions
- manage sessions
- connect to databases.
- Generate dynamic HTML.

For example, an XML file containing student information be transformed into an HTML table using XSLT.

PURPOSE: XSLT separates XML data from its presentation and converts structured XML data into a user-friendly format.

e) XML Schema

XML Schema, commonly called XSD

It can specify:

- Which elements are allowed
- The order of elements.
- Data types.
- Required and optional elements.
- Restrictions on values.

PURPOSE: It validates whether a document follows the required structure and data type.

2 How These Technologies are the Inter connected.

user → Browser → Controller → JSP/P
Business logic → Database → XML/Pars
XSLT → View → Browser.

STEP 1: XML stores structured data

```
<Student>  
  <id>101</id>  
  <name>Pradeep</name>  
  <course>CSE</course>  
</Student>
```

STEP 2:

XML schema validates the
The XML document validated element
XSD file.

For example, id can be defined
in integer name string.

STEP: 3.

Parser reads the XML.

A XML Parser loads of XML documents well-formed. The applications XML data for further processing.

STEP 4: XSLT transforms XML.

If XML data needs to be displayed browser, XSLT can transform into HTML.

XML data needs to display

XML data ~~+~~ XSLT $\rightarrow$ HTML output

Thus, XML handles the data while XSLT handles its transformation / processing.

STEPS:-

JSP and PHP Perform server side processing

JSP and PHP can receive requests from user and processing them on the server.

for example, when user submits login form.

MVC COMPONENT

ROLE

Model

Handles business logic and data operations

View

Display information using JSP/HTML.

XML Processing diagram:

